ROMAN NALAPKO

PYTHON BACKEND & APPLIED AI ENGINEER

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PROFESSIONAL SUMMARY

Python backend and applied AI engineer with 3+ years of project-based experience building RAG systems, asynchronous APIs, LLM integrations, and automation workflows. Develops testable services with FastAPI, PostgreSQL, SQLAlchemy, Qdrant, Docker, Cloudflare, and CI, from requirements through validation and handoff.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, FastAPI, asyncio, REST APIs, SQLAlchemy, Pydantic, Alembic, webhooks

Applied AI: RAG, hybrid retrieval, embeddings, semantic search, LLM evaluation, prompt/context engineering, OpenAI-compatible APIs, Gemini API, Whisper

Data & Cloud: PostgreSQL, Qdrant, Cloudflare Workers, D1, KV, pandas, document processing

Quality & Delivery: pytest, Ruff, Docker, Docker Compose, GitHub Actions, structured logging, Prometheus, Linux, Git

EXPERIENCE

Independent Python Backend & Applied AI Engineer | *Private Client & Project-Based Work* | *Mar 2023 – Present* | *Remote*

- Built Python backend and automation solutions for private clients, translating requirements into APIs, data flows, automated checks, Dockerized services, and handoff documentation.
- Delivered a private AI-powered personal finance assistant with Telegram as the interface; processed text and voice inputs through Gemini and Whisper, validated model output, and required user confirmation before persistence.
- Implemented PostgreSQL schemas and Alembic migrations, user-scoped queries, atomic multi-transaction writes, currency-separated summaries, XLSX export, health checks, and automated tests.

SELECTED ENGINEERING PROJECTS

AI RAG Platform | *Python, FastAPI, PostgreSQL, Qdrant, Docker* | GitHub

- Built a multi-tenant document intelligence backend with PDF/TXT ingestion, durable indexing jobs, and hybrid retrieval that combines dense vectors with PostgreSQL full-text search through Reciprocal Rank Fusion.
- Added bcrypt/JWT authentication, tenant isolation, SSE answers with source citations, conversation memory, LLM evaluation, Prometheus metrics, structured JSON logs, Alembic migrations, Docker, and CI.

SeaRisk UA | *JavaScript, PWA, public maritime data* | GitHub

- Developed a browser-only maritime risk research PWA with deterministic, explainable calculations and processing of official MAIB aggregate data.
- Kept user data local and added automated checks for accessibility, governance, research methodology, and feature behavior.

EDUCATION & CREDENTIALS

Bachelor's Program in Information Systems and Technologies (F6) | Danube Institute of National University 'Odesa Maritime Academy' | Expected 2030

Selected certifications: Database Integration and Management — Microsoft; Linux Commands & Shell Scripting Essentials V2 — IBM; Trustworthy Generative AI — Vanderbilt University

LANGUAGES

Ukrainian — Native | Russian — Native | English — Intermediate (B1)